

R code for Data Science for Beginners

Day 5: Individual Exercise

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clean up your workspace

```
rm(list=ls(all=TRUE)) # remove all the named objects visible in the environment  
cat("\014") # clean your console
```

Load the data

```
library(readr)
library(tidyverse)
```

```
-- Attaching core tidyverse packages ----- tidyverse 2.0.0 --
v dplyr      1.2.1      v purrr      1.2.2
v forcats    1.0.1      v stringr    1.6.0
v ggplot2    4.0.3      v tibble     3.3.1
v lubridate  1.9.5      v tidyr      1.3.2
-- Conflicts ----- tidyverse_conflicts() --
x dplyr::filter() masks stats::filter()
x dplyr::lag()     masks stats::lag()
i Use the conflicted package (<http://conflicted.r-lib.org/>) to force all conflicts to become
```

```
library(ggplot2)
world <- read_csv("world.csv")
```

Rows: 191 Columns: 62

```
-- Column specification -----
Delimiter: ","
chr (24): country, colony, dem_other5, democ_regime, district_size3, enpp_3,...
dbl (37): confidence, decentralization, dem_other, durable, effectiveness, f...
lgl (1): sources
```

i Use ``spec()`` to retrieve the full column specification for this data.

i Specify the column types or set ``show_col_types = FALSE`` to quiet this message.

Remember that Quarto uses a relative path so always save your data in the same folder (or under the same folder) with your Quarto code.

Democracy and female representation

Do democratic countries (`democ_regime == "Yes"`) have better female representation than non-democratic countries? Please answer this question by showing some graphs to assess the relationship between these two variables.

Initial data clean up

Analyze y

```
summary(world $ women09)
```

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.	NAs
0.00	9.70	15.55	17.18	22.95	56.30	11

Analyze x

```
summary(world $ democ_regime)
```

Length	N.unique	N.blank	Min.nchar	Max.nchar	NAs
191	2	0	2	3	2

Note: There are some missing values. Labels are not intuitive. So, we will deal with these two first.

Create a smaller data set that omits NA observations.

```
women.dem <- world[ is.na(world $ women09) == FALSE & is.na(world $ democ_regime) == FALSE, ]
```

Let's make sure that we did this correctly.

```
summary(women.dem $ women09)
```

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.00	9.85	15.55	17.23	22.73	56.30

```
summary(women.dem $ democ_regime)
```

Length	N.unique	N.blank	Min.nchar	Max.nchar
178	2	0	2	3

We can see that NA cases have been correctly removed.

Re-labeling the values

```
women.dem $ dem.dum <- factor(women.dem $ democ_regime,  
                              levels = c("Yes", "No"),  
                              labels = c("Democracy", "Autocracy"))
```

Q1: Describe Y (numerically and graphically)

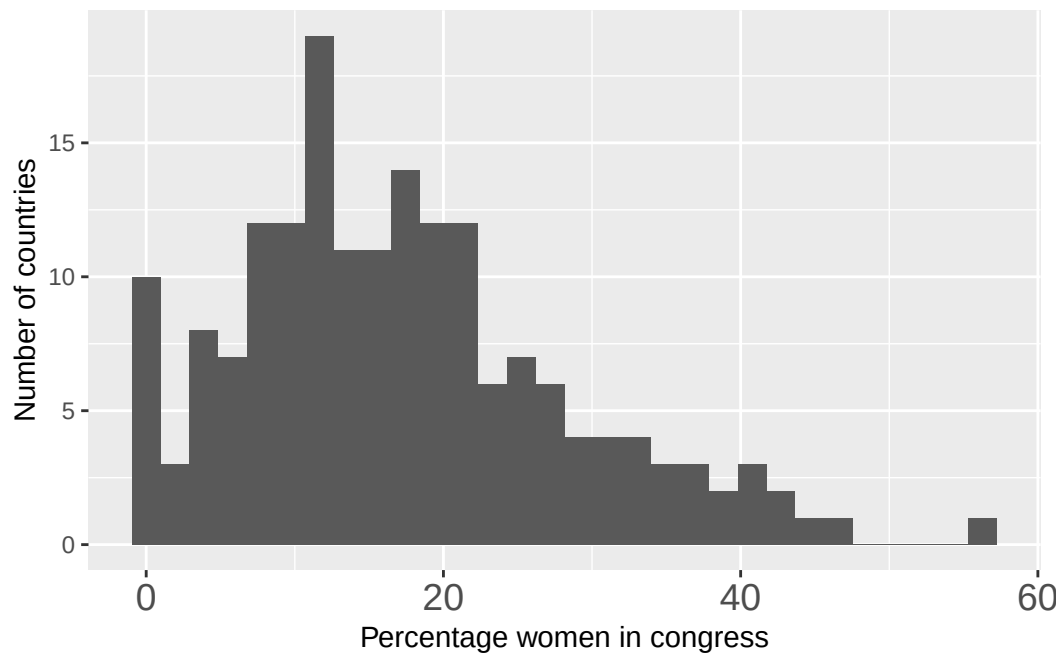
Numerical summary

```
summary(women.dem $ women09)
```

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.00	9.85	15.55	17.23	22.73	56.30

Graphical summary

```
g <- ggplot(women.dem, aes(x = women09)) + geom_histogram(bins = 30)  
g <- g + theme(axis.text.x = element_text(size = 14)) # specify text font axis.text.x to 14  
g <- g + xlab("Percentage women in congress") + ylab("Number of countries")  
g
```



Q2: Describe X (numerically and graphically)

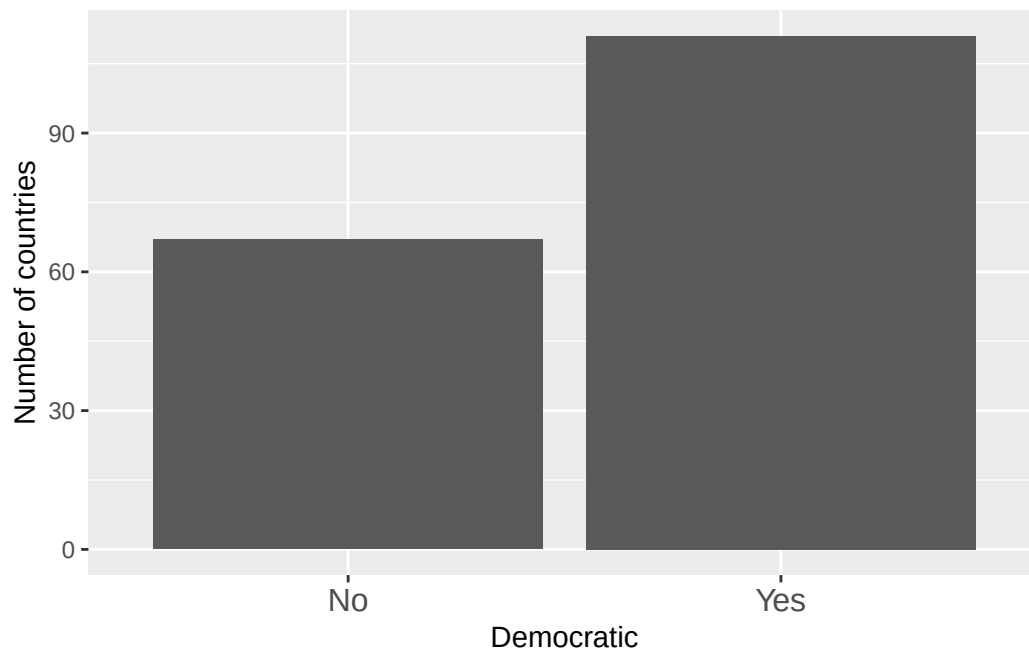
Numerical summary (frequency table)

```
table(women.dem $ democ_regime)
```

```
No Yes
67 111
```

Graphical summary (bar chart)

```
g2 <- ggplot(women.dem, aes(x = democ_regime)) + geom_bar()
g2 <- g2 + theme(axis.text.x = element_text(size = 12))
g2 <- g2 + xlab("Democratic") + ylab("Number of countries")
g2
```

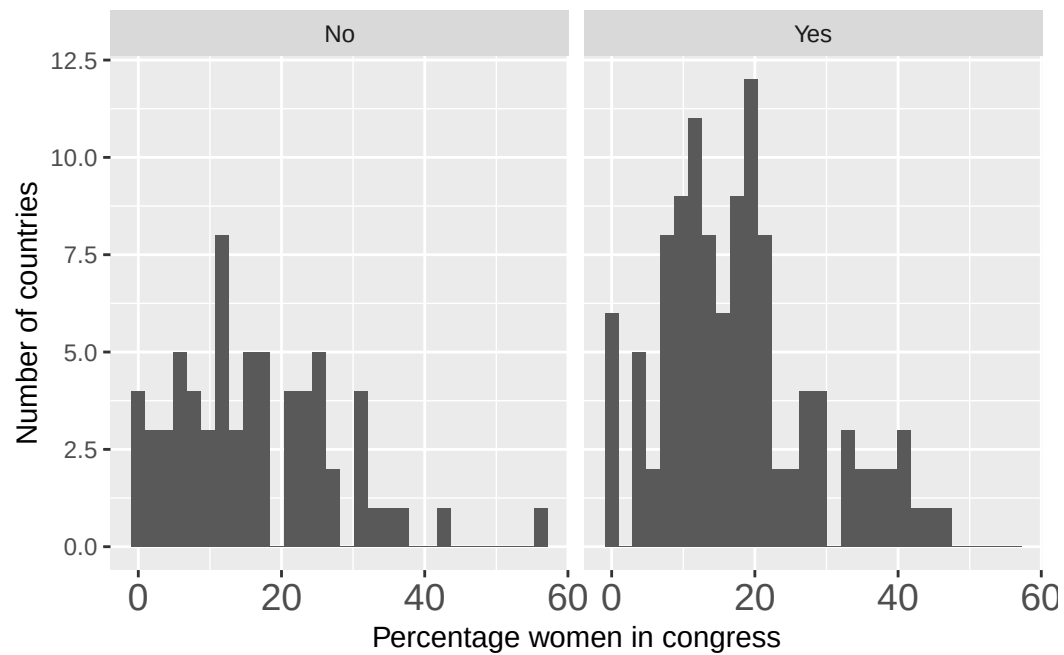


Q3: Describe X-Y graphically

Histograms

```
g3 <- ggplot(women.dem, aes(x = women09)) + geom_histogram()
g3 <- g3 + theme(axis.text.x = element_text(size = 14))
g3 <- g3 + xlab("Percentage women in congress") + ylab("Number of countries")
g3 <- g3 + facet_grid(.~ democ_regime)
g3
```

``stat_bin()`` using ``bins = 30``. Pick better value ``binwidth``.



Box-plots

```
g4 <- ggplot(women.dem, aes(y = women09, x = democ_regime)) # Remember that we need to specify y and x
g4 <- g4 + geom_boxplot()
g4 <- g4 + theme(axis.text.x = element_text(size = 14))
g4 <- g4 + xlab("Democratic") + ylab("Percentage women in congress")
g4
```

